# Simpson's Paradox — Corpus Summary

A digital mirror of Simpson's Paradox — the paradox falls out of the DAG as a derived fact, never modeled directly.

Generated 2026-07-07 from live vw_* views — same data as /conclusions. SSoT is the rulebook JSON.

## Scope boundary (conc-09)

The instrument classifies allocation geometry only. Causal interpretation is external, gated by AdjustmentAppropriate. Berkeley: highest AllocationDistortion (0.193) but CausalRole=contested — geometry and confounding are not the same thing.

## Scope & limits

Theorem conclusions follow from definitions (regression checks for the transpiler). Domain conclusions and corpus discovery findings are statistics on a curated convenience sample — directional, not inferential. See conc-14 before treating cross-domain patterns as laws.

Allocation sweep: 10 steps per study, f "  [0.05, 0.95] (SweepStudyConfig). Latent flip = Type D at observed allocation + PooledGapCrossesZero somewhere in that range — not necessarily at observed allocation.

## Epistemic summary (tiered)

Proved by construction (theorem): 1
Established in instrument (instrument · taxonomy · methodology · scope): 16
Observed in this corpus (domain — provisional): 4
Total conclusions: 21
Corpus hypotheses (loop-61): 11 / 16 confirmed
Consistency checks (definition-linked): 1 / 1 pass
Latent Type-D (under sweep contract above): 51 / 70 (72.9%)
Studies in corpus: 104
Distortion taxonomy: A 9 · B 11 · C+ 8 · C"  6 · D 70

## Pre-registered discovery (loop-61)

Consistency checks — confirm implementation matches stated algebra (same tier as conc-12 / inv-signal-purity-sign-flip). Not independent discoveries about nature.

### H-purity — PASS (consistency check)

Every sign-flip study has SignalPurity strictly below 0.5.

Expected: SignFlipSignalPurityMax < 0.5
Observed: maxPurity=0.483

Corpus hypotheses — contingent patterns on this convenience sample. Directional inequalities only; not inferential. See conc-14.

### H-catalog-exact-match — PASS (corpus hypothesis)

Catalog ExpectedDistortionType exact-match rate is below 50% — curator tags behave as mechanism priors, not taxonomy predictors.

Expected: TypePredictionMatchRate < 0.5
Observed: exactRate=0.198 (20/101)

### H-catalog-flip-prediction — PASS (corpus hypothesis)

Among catalog rows tagged Expected A"*B, observed type is A"*B less than 50% of the time — even sign-flip expectation fails often.

Expected: SignFlipPredictionMatchRate < 0.5
Observed: flipPredRate=0.253 note=exact=20/101 (19.800%); flipPred=20/79 (25.300%)

### H-causal-latent — PASS (corpus hypothesis)

Collider/selection studies produce latent-only flips under allocation sweep without manifest sign-flips at observed allocation.

Expected: ColliderSelectionManifestCount = 0 AND ColliderSelectionLatentOnlyCount >= 5
Observed: collManifest=0 collLatent=6 collN=8

### H-causal-manifest — PASS (corpus hypothesis)

Confounder-labeled studies produce manifest sign-flips at observed allocation (IsSignFlip=TRUE) far more often than collider/selection studies — reversal mode differs by mechanism class.

Expected: ConfounderSignFlipCount > ColliderSelectionManifestCount AND ConfounderSignFlipCount >= 10
Observed: confFlip=17 collManifest=0

### H-collider-no-manifest-v2 — FAIL (corpus hypothesis)

Collider and selection studies produce zero manifest sign-flips (DistortionType A or B) at observed allocation — including hernandez-diaz birth-weight collider tagged Expected A in catalog.

Expected: ColliderSelectionManifestCount = 0

### H-cplus-magnitude — FAIL (corpus hypothesis)

C+ studies carry higher mean AllocationDistortion than C- studies and Type-D studies — magnitude inflation is the largest quantitative error class and invisible to is_sign_flip.

Expected: CPlusAvgDistortion > CMinusAvgDistortion AND CPlusAvgDistortion > TypeDAvgDistortion

### H-domain-dist — PASS (corpus hypothesis)

Epidemiology studies carry higher mean AllocationDistortion than education studies.

Expected: EpidemiologyAvgDistortion > EducationAvgDistortion
Observed: epi=0.047 edu=0.004

### H-domain-flip-geometry-controlled — PASS (corpus hypothesis)

Economics zero-flip pattern persists among high max_stratum_imbalance studies while epidemiology/legal/sports remain elevated — domain heterogeneity survives geometry control.

Expected: EconomicsHighImbalanceSignFlipCount = 0 AND EpidemiologyHighImbalanceSignFlipRate > 0.15
Observed: econHighImbFlips=0 epiHighImbRate=0.261 legalHighImbRate=0.375 sportsHighImbRate=0.333 threshold=0

### H-domain-profiles-stable — FAIL (corpus hypothesis)

After expansion wave 2 (N>110 real studies), education and sports remain majority latent-fragile (>50% Type-D with LatentFlipPotential) while economics manifest sign-flip rate stays below 5%.

Expected: EducationLatentFraction > 0.5 AND SportsLatentFraction > 0.5 AND EconomicsSignFlipRate < 0.05

### H-econ-encoding-selection — FAIL (corpus hypothesis)

Among newly encoded economics catalog rows tagged Expected A, observed DistortionType is D more than half the time — economics robustness reflects 2×K encoding limits or catalog mechanism priors, not domain immunity to paradox.

Expected: EconomicsExpectedAMismatchRate > 0.5

### H-econ-zero — PASS (corpus hypothesis)

Economics-domain studies produce zero sign-flips in 2×K encoding.

Expected: EconomicsSignFlipCount = 0
Observed: flips=0

### H-explained-confounder — PASS (corpus hypothesis)

IsParadoxExplained is TRUE for all and only confounder sign-flip studies — adjust-resolvable confounding is the unique condition that makes a reversal fully explained.

Expected: ExplainedConfounderCount = SignFlipCount AND ExplainedNonConfounderCount = 0
Observed: ExplainedConfounderCount=17 ConfounderSignFlipCount=17

### H-latent-d — PASS (corpus hypothesis)

More than half of Type-D (SAFE) studies would flip their pooled winner under counterfactual allocation reweighting.

Expected: LatentTypeDFraction > 0.5
Observed: fraction=0.729

### H-small-effect — PASS (corpus hypothesis)

Allocation-stable Type-D studies carry larger observed pooled gaps than latent Type-D studies.

Expected: AvgPooledGapStableD > AvgPooledGapLatentD
Observed: stable=0.102 latent=0.022

### H-ultra-fragile — FAIL (corpus hypothesis)

At least four Type-D studies are sweep-fragile: signal_purity=1, allocation_distortion=0, sweep_pooled_gap_range>0.3, allocation_fragility>=10 — perfectly safe by pooled metrics yet latent-flip under reweighting.

Expected: SweepFragileCount >= 4

### H-unexplained-nonconfounder — PASS (corpus hypothesis)

Studies with IsSignFlip=TRUE and CausalRole != confounder (contested or mediator) have IsParadoxExplained=FALSE.

Expected: ContestedOrMediatorExplainedCount = 0
Observed: ContestedOrMediatorExplainedCount=0

## Proved (by construction) (1)

### conc-12-signal-purity-theorem · Proved (by construction)

SignalPurity < 0.5 is a necessary condition for allocation-driven reversal

Witnessed across all reversal studies (type A + B) in the 90+ study corpus: every study with AllocationDirection='reversal' has SignalPurity < 0.5. The algebraic proof: reversal requires SignedPooledGap and CorrectedGap to have opposite signs, which requires AllocationDistortion > |CorrectedGap|, which is exactly SignalPurity < 0.5. inv-signal-purity-sign-flip enforces this as a DAG invariant. AvgSignalPurityReversal "H 0.35 vs AvgSignalPurityNonReversal "H 0.65 — gap of ~0.30.

Loop replay: loop-44 (5a14dcb, Sun Jun 28 2026 00:00:00 GMT-0500 (Central Daylight Time))
Protecting invariants: 1

## Established in instrument (9)

### conc-03-binary-excludes-partial · Established in instrument

The unanimity criterion (IsReversal) incorrectly excludes Berkeley 1973 despite its being a Simpson-type paradox

Berkeley: WSGSUM=" 0.0642, SignedPooledGap=+0.1292, IsSignFlip=TRUE, AllocationDistortion=0.1934 — greater distortion than kidney-1986. But IsReversal=FALSE because dept-C marginally favors male. The unanimity criterion is a strict subtype, not the primary definition.

Loop replay: loop-17 (ca31095, Sat Jun 27 2026 00:00:00 GMT-0500 (Central Daylight Time))
Protecting invariants: 4

### conc-04-type-c-exists · Established in instrument

Allocation can compress the pooled signal without flipping it (type-C): a class invisible to IsReversal and IsSignFlip but visible to AllocationDistortion

compressed-synthetic: A wins every stratum and wins pooled, but the pooled margin is 6pp rather than the equal-weight 10pp. AllocationDistortion=0.0400, IsSignFlip=FALSE. The pooled winner is correct; the effect size is wrong by 4pp.

Loop replay: loop-19 (ca31095, Sat Jun 27 2026 00:00:00 GMT-0500 (Central Daylight Time))
Protecting invariants: 4

**conc-05-real-data-generalizes · Established in instrument**

Two real published studies (reintjes-2000, radelet-1981) hydrate as type-A full reversals without schema changes

Both studies entered via 05b-customize-data.sql. Witnessed: reintjes AllocationDistortion=0.0516, DistortionType=A; radelet AllocationDistortion=0.0684, DistortionType=A. No DAG or schema change was needed — the hydration contract generalizes.

Loop replay: [loop-20b (ca31095, Sat Jun 27 2026 00:00:00 GMT-0500 (Central Daylight Time))]

**conc-06-policy-derivable · Established in instrument**

PolicyImplication is domain-neutral: the same formula produces researcher actions across medicine, law, and education

All five current studies (kidney-1986, berkeley-1973, compressed-synthetic, kidney-balanced, balanced-synthetic) plus two real studies produce unambiguous PolicyImplication values from DistortionType alone. No domain-specific knowledge is encoded in the formula.

Loop replay: [loop-21 (ca31095, Sat Jun 27 2026 00:00:00 GMT-0500 (Central Daylight Time))]

**conc-07-model-self-portrait · Established in instrument**

ModelSummary witnesses the instrument's own epistemic coverage as a first-class derived fact

ModelSummary aggregates ReversalCount, ExplainedCount, TypeA/B/C/D counts, and AvgAllocationDistortion across all studies. The model describes its own completeness in a single row.

Loop replay: [loop-10 (b96e0fe, Sat Jun 27 2026 00:00:00 GMT-0500 (Central Daylight Time))]

**conc-11-reversal-recovery · Established in instrument**

The model can produce the allocation-corrected winner directly from existing DAG fields — no new data required

CorrectedGap = WeightedStratumGapSum (already derived in loop-12/17). CorrectedWinner is a sign-read on CorrectedGap. CorrectedVsPooledAgreement is TRUE for Type-D studies and FALSE for Type-A/B studies. For kidney-1986: CorrectedGap=+0.0537, CorrectedWinner=A, CorrectedVsPooledAgreement=FALSE (pooled said B; corrected says A). The reversal recovery is free — it costs zero new data.

Loop replay: [loop-27 (ca31095, Sat Jun 27 2026 00:00:00 GMT-0500 (Central Daylight Time))]
Protecting invariants: 3

**conc-15-latent-flip-potential · Established in instrument**

Type-D at observed allocation does not imply allocation-stable — 70% of SAFE studies flip under counterfactual reweighting

Sensitivity over sweep range f "  [0.05, 0.95]; not a claim about realistic reallocation alone.

LatentTypeDCount=45, TypeDCount=64, LatentTypeDFraction=0.703. LatentFlipPotential=TRUE when DistortionType=D AND PooledGapCrossesZero=TRUE. Examples: uc-systemwide-admissions (Type D, sweep range 0.56), recsys-offline-eval (Type D, sweep range 0.56). The ScreeningTier SAFE label describes observed allocation only; sweep reveals latent flip potential.

Loop replay: [loop-61 (7b3938e, Mon Jun 29 2026 00:00:00 GMT-0500 (Central Daylight Time))]
Protecting invariants: 1

**conc-16-small-effect-fragile · Established in instrument**

In this corpus, allocation-stable Type-D rows carry larger observed pooled gaps than latent Type-D rows

AvgPooledGapStableD=0.102 across 19 stable Type-D studies; AvgPooledGapLatentD=0.022 across 45 latent Type-D studies. Stable studies also have lower sweep range (avg 0.072 vs 0.211). Corpus-descriptive pattern on loop-61 convenience sample — not a deductive law. Small observed effects tend to be allocation-fragile even when currently classified SAFE.

Loop replay: [loop-61 (7b3938e, Mon Jun 29 2026 00:00:00 GMT-0500 (Central Daylight Time))]

**conc-18-causal-role-flip-mode · Established in instrument**

Causal role predicts manifest vs latent reversal mode — confounders flip at observed allocation; collider/selection flips stay sweep-latent

Loop-62 research sweep: H-causal-manifest PASS (confFlip=15 vs collManifest=0). H-causal-latent PASS (collManifest=0, collLatent=5, collN=7). Confounder Type-D rows often show latent-only sweep sensitivity (45 latent-only) without manifest sign-flip — the mechanism split is about reversal mode when sign-flips occur, not absence of sweep fragility. Existing LatentFlipPotential + StratumCausalRole suffice; defer ManifestReversal first-class fields to loop-67+.

Loop replay: loop-62

## Taxonomy (1)

### conc-13-cplus-cminus-opposite-errors · Taxonomy

C+ (amplification) and C- (compression) are epistemically opposite errors with opposite clinical implications

titanic-1912 (C+, DistortionRatio=1.359): allocation inflates first-class survival advantage from 27.5pp corrected to 37.4pp pooled — overconfidence. melanoma-altman-1991 (C+, DistortionRatio=2.087): allocation doubles apparent female survival advantage — severe overconfidence. hannan-1994 (C-, DistortionRatio=0.314): allocation compresses CABG advantage from 8.9pp corrected to 2.8pp pooled — underconfidence, risk of premature treatment abandonment. The prior type-C category hid this distinction. Now separated b…

Loop replay: loop-43 (5a14dcb, Sun Jun 28 2026 00:00:00 GMT-0500 (Central Daylight Time))

## Methodology (4)

### conc-10-instrument-specifiable · Methodology

The full instrument can be specified precisely enough for any researcher to apply it to a new study

Witnessed in loop-26: InstrumentSpec table enumerates 5 input fields, 10 derived coordinates, 2 classification rules, and 3 adapter-requirement rows. Machine-verifiable against the full 90+ study corpus.

Loop replay: loop-26 (ca31095, Sat Jun 27 2026 00:00:00 GMT-0500 (Central Daylight Time))

### conc-14-scale-is-discovery-path · Methodology

Cross-domain SignalPurity correlations require a pre-registered unbiased corpus of 50+ studies to distinguish pattern from selection artifact

Partial domain patterns witnessed across 90+ studies (loop-61): epidemiology avg distortion 0.050 vs education 0.004; legal/sports/epidemiology flip rates ~27% vs economics 0%. Full causal domain!'DistortionType mapping still requires pre-registered expansion beyond convenience sample, but the instrument now supports DiscoveryHypotheses/Findings for ongoing corpus-scale tests.

Loop replay: loop-61 (7b3938e, Mon Jun 29 2026 00:00:00 GMT-0500 (Central Daylight Time))

### conc-21-catalog-tags-not-predictors · Methodology

Catalog ExpectedDistortionType tags are mechanism priors, not taxonomy predictors — exact match rate 19.4%

Loop-65 audit: 18/93 exact matches (19.4%), 75 mismatches. Sign-flip prediction 18/71 (25.4%). Biggest bucket Expected A!'observed D (23 studies). Both H-catalog hypotheses PASS (<50%). Defer ExpectedSignFlip vs ExpectedDistortionType split to loop-67+.

Loop replay: loop-65

### conc-23-research-sweep-synthesis · Methodology

Research sweep loops 62–66: confirmed laws vs corpus patterns vs deferred vocabulary

Laws (build-breaking): SignalPurity<0.5 for sign-flips (loop-64), IsParadoxExplained!"confounder (loop-63). Corpus patterns: causal-role reversal mode (loop-62), latent Type-D 70% (loop-61), catalog tag falsification 19% exact (loop-65), domain gap survives geometry control (loop-66). Deferred loop-67+: ManifestReversal/LatentReversal fields, ExpectedSignFlip catalog split, Discovery UI polish. inv-discovery-all-confirmed: 11/11 PASS.

Loop replay: loop-66
Protecting invariants: 1

## Scope / open question (2)

**conc-08-type-c-real · Scope / open question**

Does the published literature contain real type-C studies (compression without sign flip)?

Witnessed in loop-24: hannan-1994 (CABG surgery), titanic-1912, melanoma-altman-1991, and coffee-tverdal-2020 all hydrate as DistortionType=C. AllocationDistortion>0.01, IsSignFlip=FALSE — the correct pooled direction but compressed magnitude. The type-C cell is not a synthetic pathology.

Loop replay: [loop-24 (ca31095, Sat Jun 27 2026 00:00:00 GMT-0500 (Central Daylight Time))]()

**conc-09-causal-vs-geometric · Scope / open question**

AllocationDistortion is a geometric measure; causal interpretation requires external domain knowledge, gated by AdjustmentAppropriate

Berkeley has the highest AllocationDistortion (0.1934) but CausalRole=contested (mediator vs confounder), proving that geometric severity and causal confounding are not the same thing. The instrument is deliberately geometric: it classifies distortion type and flags the corrected winner from allocation arithmetic alone. Whether it is safe to act on that correction as a causal claim is answered by AdjustmentAppropriate — which gates on ConditioningRisk and CausalClaimStatus. The two layers are complementary, not …

Loop replay: [loop-30 (ca31095, Sat Jun 27 2026 00:00:00 GMT-0500 (Central Daylight Time))]()
Protecting invariants: 1

## Observed in this corpus (4)

**conc-01-paradox-is-derived · Observed in this corpus**

Simpson's Paradox is a derived fact from the DAG, not a modeled entity

IsReversal (and later IsSignFlip) falls out as a calculated field on TreatmentRankings, derived from aggregated CaseCell counts. No 'ReversalDetection' entity was ever needed.

Loop replay: [loop-04 (b96e0fe, Sat Jun 27 2026 00:00:00 GMT-0500 (Central Daylight Time))]()
Protecting invariants: 2

**conc-02-reversal-from-allocation · Observed in this corpus**

The reversal in kidney-1986 is entirely an allocation artifact — same rates with balanced allocation produce no reversal

kidney-balanced uses identical stratum success rates to kidney-1986 but with equal allocation (175 cases per cell). IsReversal=FALSE, AllocationDistortion=0. The paradox is algebraically isolated as a property of allocation, not of treatment efficacy.

Loop replay: [loop-14 (49f01e7, Sat Jun 27 2026 00:00:00 GMT-0500 (Central Daylight Time))]()
Protecting invariants: 3

**conc-17-economics-flip-free · Observed in this corpus**

Economics-domain 2×K encodings in this corpus produce zero sign-flips — composition studies read as Type D or C''

This corpus only (n=8 economics studies); see conc-14 for inferential limits.

EconomicsSignFlipCount=0 across 8 economics studies (6 Type D, 2 Type C'' ). Contrast epidemiology (6 sign-flips / 22 studies, 27.3%). Composition and wage-decomposition paradoxes compress or stabilize rather than flip in this instrument's 2×K encoding.

Loop replay: [loop-61 (7b3938e, Mon Jun 29 2026 00:00:00 GMT-0500 (Central Daylight Time))]()

**conc-22-domain-flip-geometry · Observed in this corpus**

Domain flip-rate heterogeneity survives max_stratum_imbalance conditioning — economics 0%, epi/legal/sports ~27%

Corpus-specific pattern; see conc-14.

Loop-66: H-domain-flip-geometry-controlled PASS. economics_high_imbalance_sign_flip_count=0; epidemiology_high_imbalance_sign_flip_rate=0.273; legal=0.286; sports=0.273. Pattern survives geometry control — not purely catalog selection artifact.

Loop replay: loop-66

## Invariant health

Critical invariant failures: 0

inv-pooled-gap-wanders: Pooled gap should move under allocation reweighting for reversal studies. Warning-only: small-magnitude reversals may have narrow sweep range. (fail=1)

## Most misleading studies (by paradox strength)

COVID-19 CFR: Italy vs China by Age (von Kügelgen 2021) — type C+, strength 0.2743, distortion 0.529, purity 3.6% (CAUTION)
Triathlon On-Time Finish: Sex by Age Cohort — type B, strength 0.1455, distortion 0.245, purity 28.9% (DANGER)
Smoking and Mortality by Age (Cochran / Surgeon General 1964) — type B, strength 0.1091, distortion 0.167, purity 25.7% (DANGER)
UC Berkeley Graduate Admissions (Bickel et al. 1975) — contested confounder — type B, strength 0.0969, distortion 0.193, purity 24.9% (DANGER)
UC Berkeley Admissions: Six Largest Departments (Bickel et al. 1975) — type B, strength 0.0944, distortion 0.184, purity 18.8% (DANGER)
Exercise and Cholesterol Control by Age Group (Framingham-style) — type D, strength 0.0800, distortion 0.008, purity 95.2% (SAFE)
Baseball Batting Averages: Lefebvre vs Fairly World Series (Day 1994 / baseball paradox canon) — type B, strength 0.0606, distortion 0.082, purity 20.9% (DANGER)
Whickham Smoking Cohort — 20-year Mortality by Smoking Status (Appleton et al. 1996) — type B, strength 0.0538, distortion 0.119, purity 26.9% (DANGER)

## Rulebook & repo (full derivation)

Formulas, Loops build history, InvariantChecks, and Conclusions rows live in the rulebook JSON. README.md carries the full framing.

Project folder: https://github.com/EffortlessAPI/effortless-rulebooks/tree/main/rulebook-examples/simpsons-paradox
README (full repo guide): README.md
Rulebook SSoT (formulas, Loops, InvariantChecks, Conclusions): effortless-rulebook/simpsons-paradox-rulebook.json
Rulespeak (plain-English business rules): rulespeak/rulespeak.md
Postgres substrate (views, init-db.sh): effortless-postgres/README.md
Standalone HTML explorer: simpsons-paradox-explorer.html
Live API (with ./start.sh): GET localhost:3001/api/invariant-checks · /api/treatment-rankings · /api/model-summary

Loop landing tags (replay checkpoints):

sp-loop-milestone-01-loops-01-10: conc-01-paradox-is-derived
sp-loop-milestone-03-loops-11-15: conc-02-reversal-from-allocation
sp-loop-milestone-04-loops-16-31: conc-03-binary-excludes-partial
sp-loop-milestone-04-loops-16-31: conc-04-type-c-exists
sp-loop-milestone-04-loops-16-31: conc-05-real-data-generalizes
sp-loop-milestone-04-loops-16-31: conc-06-policy-derivable
sp-loop-milestone-01-loops-01-10: conc-07-model-self-portrait
sp-loop-milestone-04-loops-16-31: conc-08-type-c-real
sp-loop-milestone-04-loops-16-31: conc-09-causal-vs-geometric
sp-loop-milestone-04-loops-16-31: conc-10-instrument-specifiable